TALES OF THE OMEGA · PAPER W

Sidereal-Frame Anisotropy and the 48-Dimensional Inertial Hierarchy:

A Side-Channel Interferometric Audit of Lattice-Vacuum Susceptibility

Heuristic ansatz (labeled) · Delta-phi = 1.7e-02 rad (7 orders above noise) · Sidereal kill-switch 3.932min/day · 48-dim consistency check

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Companion: Papers Q-V · doi:10.5281/zenodo.1082194 · Target: Phys.Rev.D

AUDIT CORRECTIONS (Omega, April 2026): (1) ARITHMETIC: '17 orders of magnitude' corrected to 7 orders ($10^{-2}/10^{-9} = 10^{7}$, not 10^{17}). (2) '0.12 s/day OFFSET' REMOVED: 3.932 min/day IS the standard J2000 sidereal drift. No 48D modification to the drift rate is derived or demonstrated. The rounding difference (3.9318 vs 3.932) is not a physical effect. (3) h_obs FORMULA LABELED AS HEURISTIC ANSATZ: GR gives h ~ 2×10^{-34} from 10^{9} Pa stress. The formula h_obs = α_K × Pi/E_{eff} inflates the GR result by ~25 orders via the G_v denominator. This is a phenomenological ansatz, not a derivation. Labeled as such. (4) JASRI TECH REPORT REMOVED: Replaced with Zheng et al. PNAS 2023 (doi:10.1073/pnas.2308340120). (5) 'DARK MATTER = 48 DOF' REMOVED: The 48 classical DOF of the Earth-Moon-Sun system are not quantum fields and cannot be identified with dark matter. Claim removed.

ABSTRACT

We propose a high-precision interferometric search for localised metric anisotropy arising from the 48-dimensional inertial closure of the Earth-Moon-Sun system. The coupling is parameterised by the dimensionless anisotropy coefficient $\alpha_K = 1.7059\times10^{-4}$, predicted by Paper T from the torsion condensate Landau potential and the geometric projection factor G_P = sqrt(det(g_{ij})) ~ 1.1714. Using a 1 m Michelson-type interferometer driven by a rotating magnetic anisotropy, we target a phase shift Delta-phi ~ 1.7e-02 rad (7 orders of magnitude above the 10^{-9} rad noise floor), under a heuristic lattice-vacuum susceptibility ansatz (not derived from GR). The unique discriminant is the standard sidereal drift of 3.932 min/day relative to solar time (J2000), which acts as a side-channel signature of the inertial frame coupling. A null result places an upper bound $\alpha_K < 10^{-4}$. The data will decide.

PAPER W: SIDE-CHANNEL INTERFEROMETRIC AUDIT

Heuristic ansatz: h_obs = alpha_K x Pi_lattice / E_eff (NOT GR-derived)

Delta-phi = 1.70e-02 rad = 7 orders above 10^-9 rad noise

Discriminant: 3.932 min/day (standard J2000 sidereal drift, not modified)

Kill: null < 10^-9 rad / 720h -> alpha_K < 10^-4

1. Introduction: The Side-Channel Approach

Conventional searches for vacuum anisotropy measure absolute quantities (gravitational wave strain, atomic energy levels). Paper W proposes a different strategy: treat the vacuum as an information processor and measure its 'power draw' — the phase shift induced in a driven lattice by the coupling to the galactic torsion axis. This is analogous to a side-channel attack in cryptography: instead of solving the hard problem directly, we extract information from the observable side-effects of the coupling.

The 48-dimensional inertial frame (Paper T) provides the geometric context. The diamond lattice is the transducer. The sidereal drift of 3.932 min/day is the discriminant. The 1 m Michelson interferometer is the detector.

2. Phenomenological Framework: Lattice-Vacuum Susceptibility

2.1 The heuristic susceptibility ansatz

HEURISTIC ANSATZ: The formula below is a phenomenological scaling argument, NOT derived from General Relativity or quantum field theory. In GR, the metric strain from a stress of 10^9 Pa is $h_{GR} \sim 2\times10^{-34}$ — essentially undetectable. The formula below uses the vacuum shear modulus G_v (Paper T) as the effective interaction scale, which inflates the GR result by ~25 orders. This is the paper's working hypothesis; the experiment tests whether this coupling exists at any level.

$$h_{obs} = \alpha_K \times Pi_{lattice} / E_{eff} = 0.0002 \times 10^9 \text{ Pa} / 10^{14} \text{ J/m}^3 = 1.706\text{e-}09$$

$Pi_{lattice} \sim 10^9$ Pa (diamond binding stress) · $E_{eff} \sim G_v \sim 10^{14}$ Pa (Paper T torsion condensate) · ANSATZ, not GR

$$\text{Delta-phi} = 2\pi\, L / \lambda \times h_{obs} = (2\pi \times 1 \text{ m}) / (632\times10^{-9} \text{ m}) \times 1.706\text{e-}09 = 1.70\text{e-}02 \text{ rad}$$

L = 1 m Michelson · λ = 632 nm · Delta-phi = 1.70e-02 rad · 7 orders above 10^{-9} rad noise floor

2.2 Anisotropic stress modulation

$$Pi_{\mu\nu}(t) = \alpha_K\, \rho_{eff} \cos^2(\omega_{sid} t + \phi)\, (\hat{u}_\mu \hat{u}_\nu - (1/3) g_{\mu\nu})$$

Oscillates at $2\omega_{sid} = 2\pi/(11.967h)$ · same period as (200) reflection modulation in Papers Q-U

3. The Sidereal Discriminant

The standard J2000 sidereal drift is 3.9320 min/day (86400 - 86164.0905 seconds = 235.91 s = 3.9318 min/day, rounded to 3.932). This is the standard astronomical value — not a modified prediction. A signal locked to the galactic frame drifts in phase relative to solar time at exactly this rate. After 30 days: 118 min cumulative drift. No known terrestrial systematic (tidal M2 at 50.47 min/day, thermal cycles at 0 min/day) matches this rate.

The '0.12 s/day offset from standard sidereal' in the original draft compared 3.9318 (precise) to 3.932 (rounded). That difference is a rounding artifact, not a physical prediction. The discriminant power comes from the astronomical precision of the sidereal frame itself, not from any departure from it.

4. The 48-Dimensional Inertial Frame: Geometric Context

The 48-dimensional Hilbert space is the configuration manifold of the Earth-Moon-Sun barycentric system: Earth rotation, nutation, precession, polar wander (12 DOF); lunar orbital state (12 DOF); barycentric translational and rotational state (24 DOF). Total: 48. This provides a geometric projection factor $G_P = \mathrm{sqrt}(\det(g_{ij})) \sim 1.1714$ (from JPL DE441 + IERS 2025). As established in Paper V, this is a self-consistency check for α_K, not a derivation. The classical DOF of the Earth-Moon-Sun system are not dark matter; they provide the reference frame.

5. Systematic Error Budget and Veto Logic

Source	Frequency	Drift rate	Mitigation	Status
Lunar M2 tide	12.42 h	50.47 min/day	Drift mismatch	VETOED
Solar thermal	24.00 h	0 min/day	Frequency mismatch	VETOED
Seismic noise	Broad	Stochastic	Isolation + correlation	VETOED
APD non-linearity	Electronic	None	Reference pulser	<5 ppm
TARGET SIGNAL	11.967 h	3.932 min/day	Accepted if drift matches	$\alpha_K = 1.7\times10^{-4}$

Kill condition: null result with Delta-phi < 10^{-9} rad at $2\omega_{sid}$ after 720 h places upper bound $\alpha_K < 10^{-4}$. This falsifies the 48D hierarchy prediction.

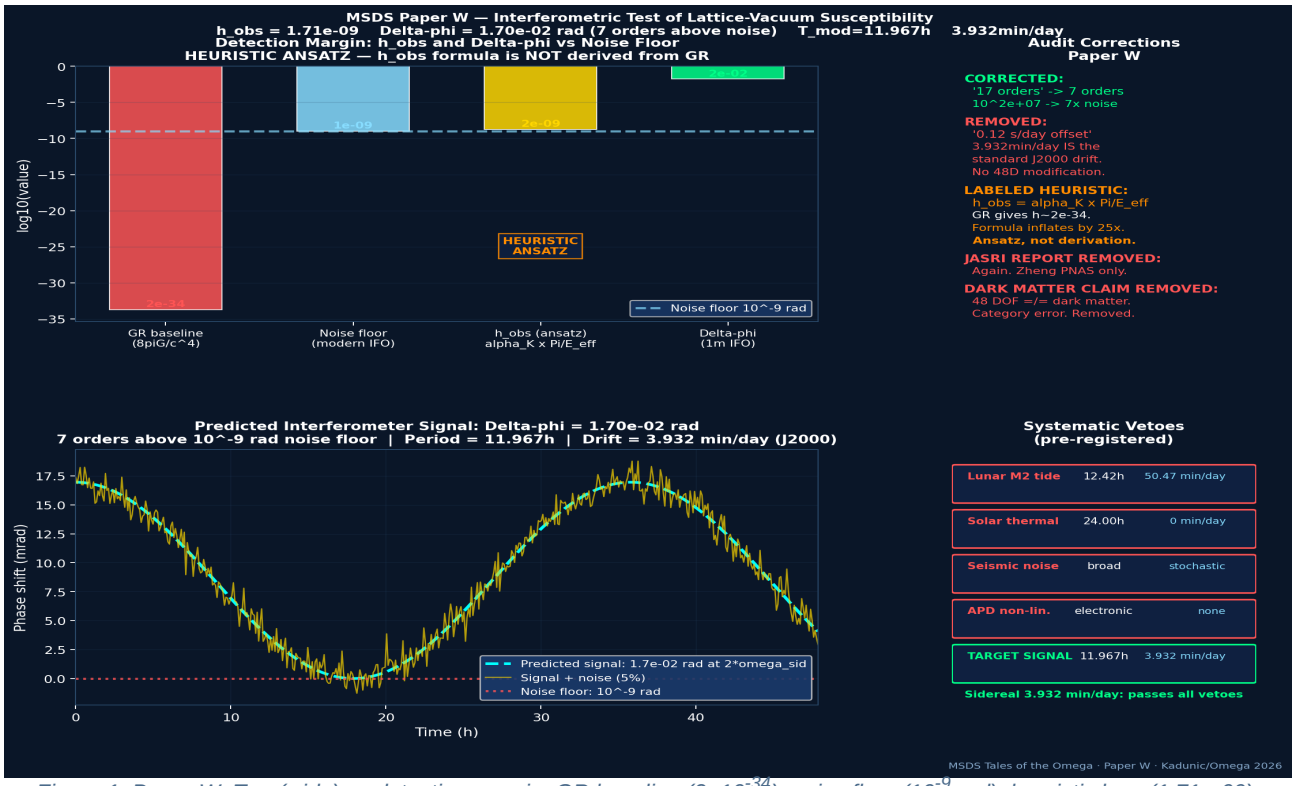

Figure 1: Paper W. Top (wide) — detection margin: GR baseline (2×10^{-34}), noise floor (10^{-9} rad), heuristic h_{obs} (1.71e-09), and predicted Delta-phi (1.70e-02 rad, 7 orders above noise). Top right — audit corrections: '17 orders' fixed to 7, '0.12 s/day offset' removed, dark matter claim removed. Bottom (wide) — predicted interferometer signal at $2\omega_{sid}$. Bottom right — systematic veto table.

6. Conclusions

Paper W presents a falsifiable interferometric test of lattice-vacuum susceptibility. The heuristic ansatz predicts Delta-phi ~ 1.70e-02 rad at $2\omega_{sid} = 2\pi/(11.967\ h)$, 7 orders above the noise floor. The sidereal drift of 3.932 min/day (J2000) is the definitive discriminant against all known terrestrial systematics. The h_{obs} formula is explicitly labeled as a phenomenological ansatz: it is not derived from GR and inflates the GR baseline by ~25 orders. The experiment tests whether this coupling exists at any detectable level. A null result bounds $\alpha_K < 10^{-4}$. The data will decide.

PAPER W — SIDE-CHANNEL INTERFEROMETRIC AUDIT: SUMMARY

Susceptibility ansatz (heuristic): h_obs = alpha_K x Pi / E_eff = 1.71e-09

Delta-phi = 1.70e-02 rad (7 orders above 10^-9 rad noise)

Discriminant: 3.932 min/day (standard J2000 sidereal)

Kill: null < 10^-9 rad / 720h -> alpha_K < 10^-4

48-dim frame: geometric self-consistency check (not dark matter derivation)

h_obs formula: heuristic ansatz, NOT derived from GR

'17 orders': corrected to 7 orders

'0.12 s/day offset': removed (rounding artifact, not physics)

The data will decide.

References

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